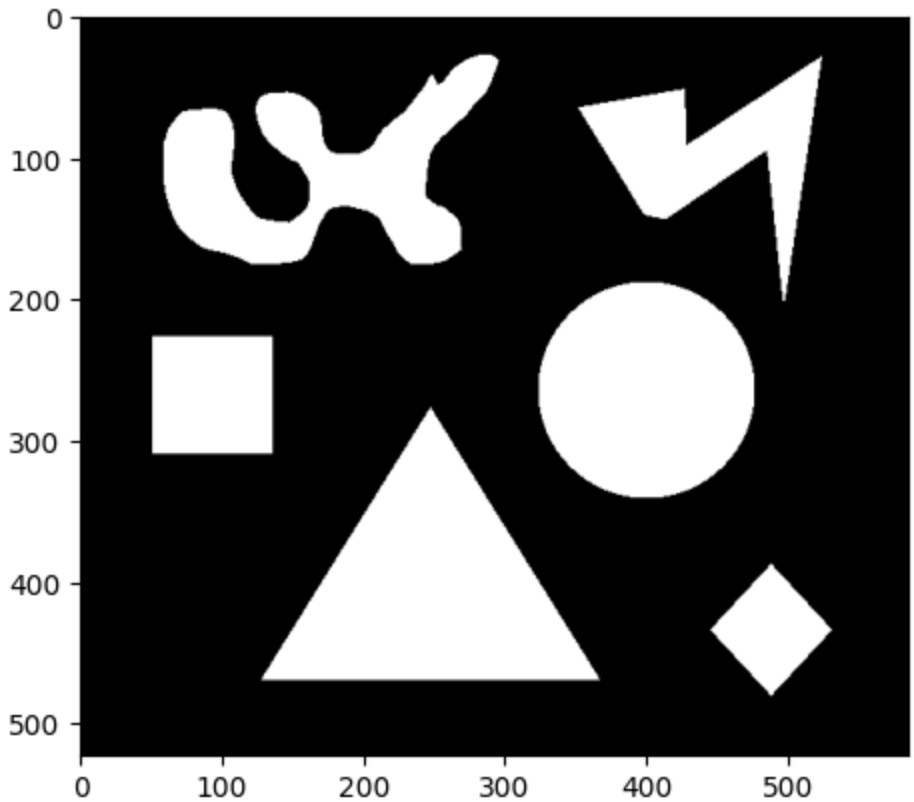
INSTITUTO SUPERIOR TECNOLÓGICO QUITO

Excelencia en Educación Superior

4.1 Descriptores de Forma

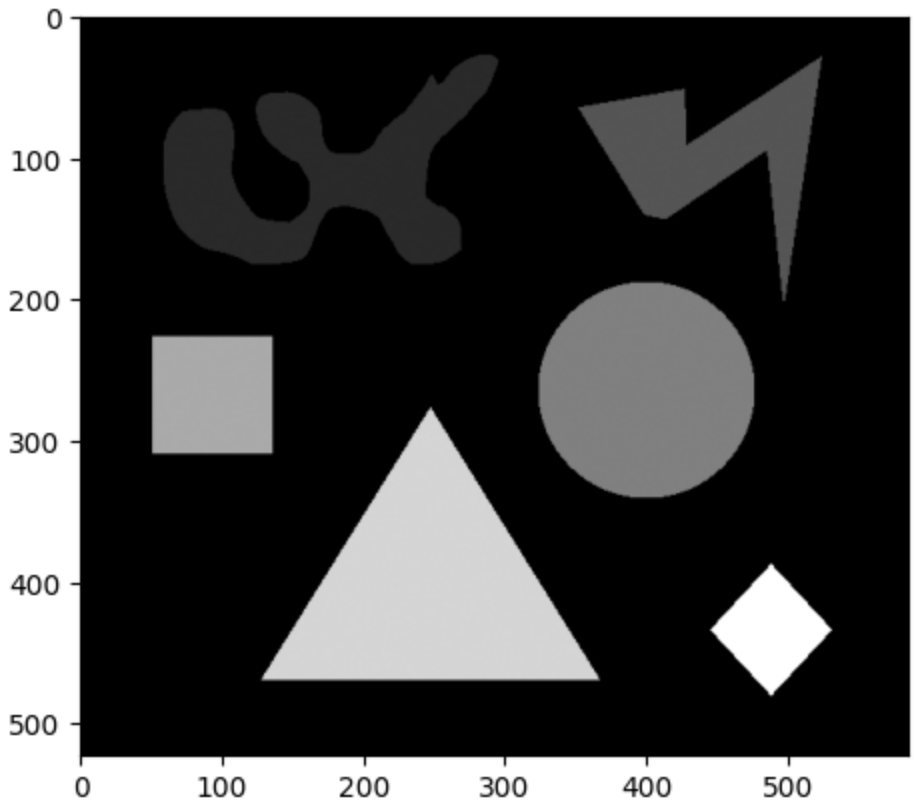
Erick Mejia

In [22]:
```
# En primer lugar, obtenemos la máscara de la imagen original mediante la umbraliza
import matplotlib.pyplot as plt
import cv2
import numpy as np
img = cv2.imread('images/figuras.png')
img_gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
_, mask = cv2.threshold(img_gray,0,1,cv2.THRESH_OTSU)

plt.imshow(mask, cmap='gray')
plt.show()
```

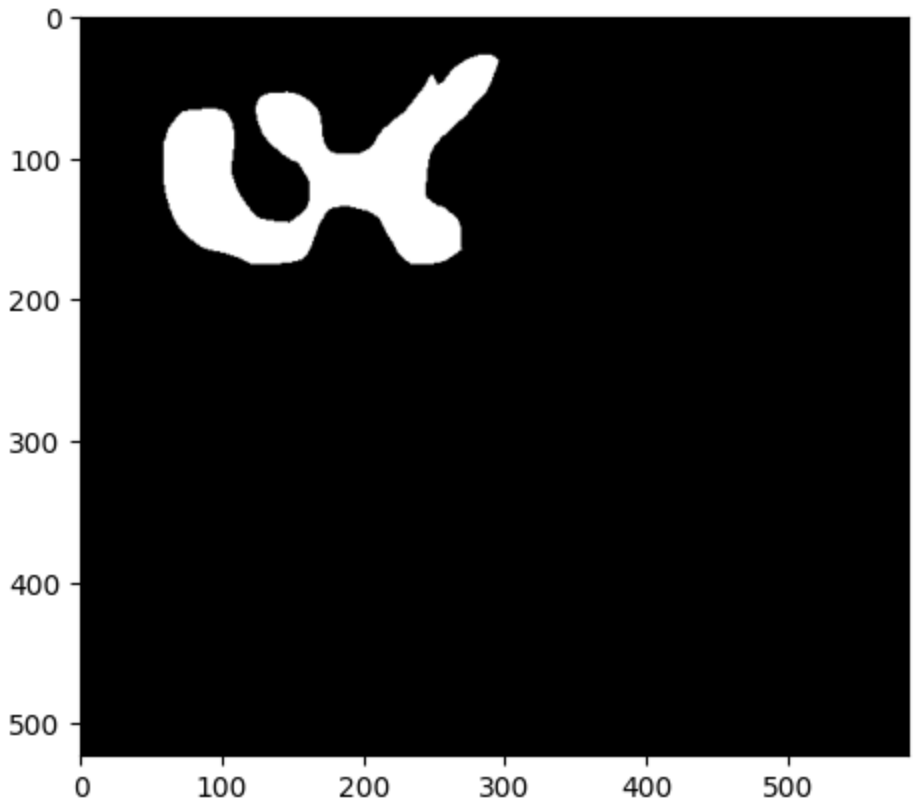


In [23]:
```
# Convertimos la imagen binaria en una imagen de etiquetas para tener acceso a cada
from skimage.measure import label

lab, num = label(mask, return_num=True)

# Importante:
# En 'num' devuelve 6 valores correspondientes a los 6 objetos que hay en la imagen
# Sin embargo, 'lab' contiene hasta 7 valores diferentes, ya que también tiene en c
print('número de objetos: ', num)
v,c = np.unique(lab, return_counts=True)
print('posibles valores de intensidad: ', v)

plt.imshow(lab, cmap='gray')
plt.show()
```

```
número de objetos:  6
posibles valores de intensidad:  [0 1 2 3 4 5 6]
```

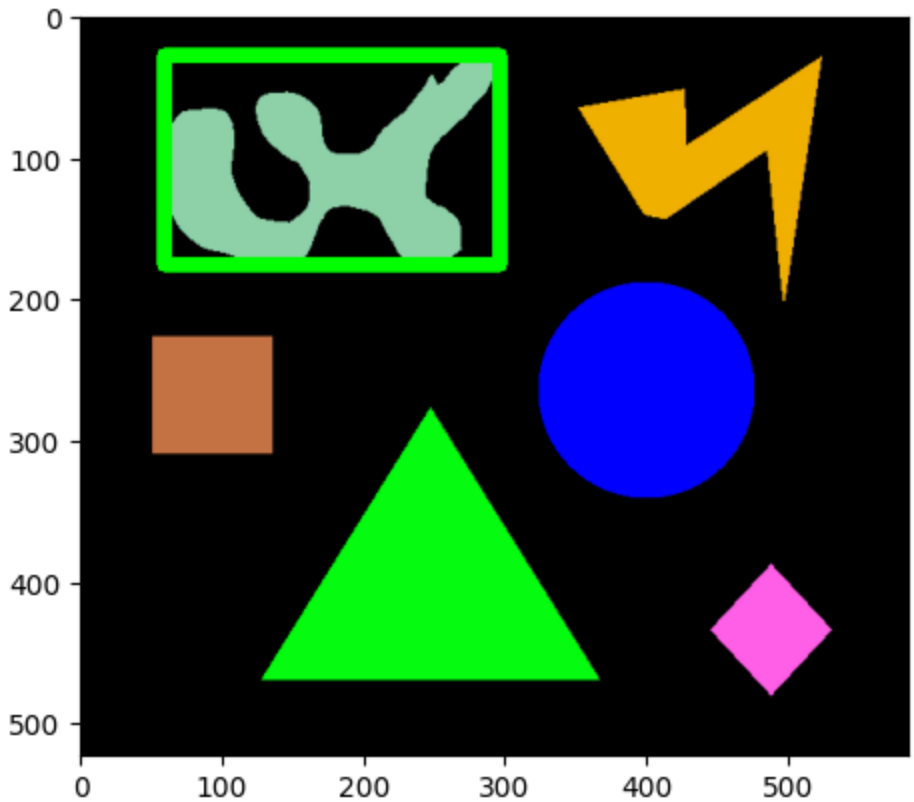


In [24]:
```
# Podemos acceder de manera independiente al objeto que nos interese a partir de su
objeto = lab == 1
objeto = objeto.astype('uint8')
plt.imshow(objeto, cmap='gray')
plt.show()
```

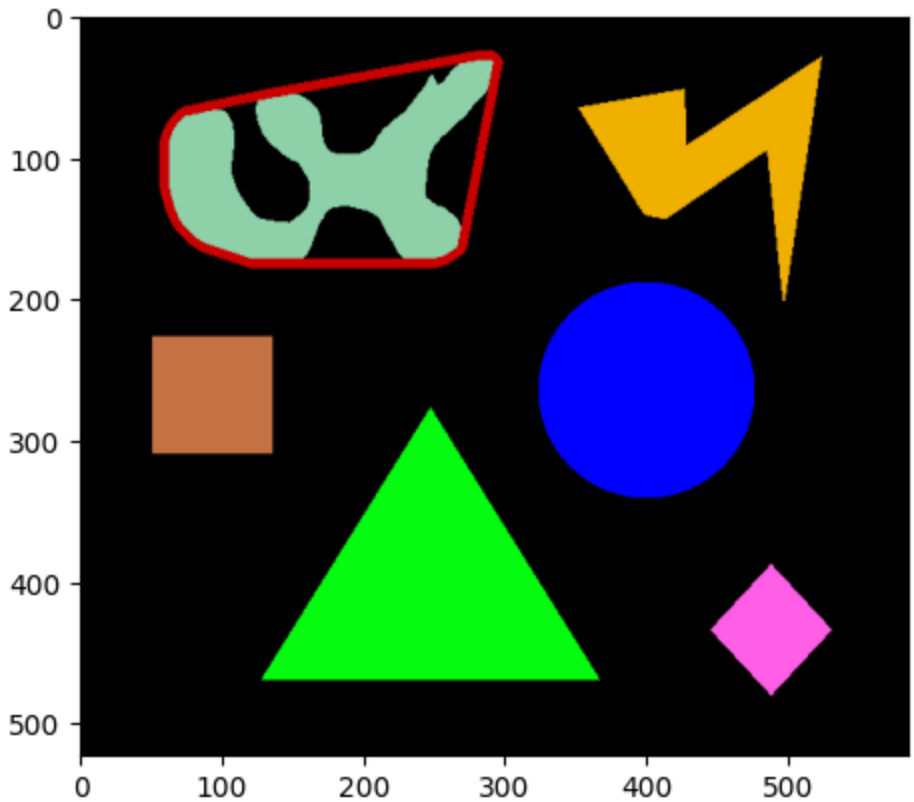


In [25]:
```
from skimage.measure import regionprops
# https://scikit-image.org/docs/dev/api/skimage.measure.html#skimage.measure.region

# Después, podemos extraer la bounding box de dicho objeto
prop = regionprops(objeto)
bbox = prop[0].bbox
print(bbox)
start_point = (bbox[1], bbox[0])
end_point = (bbox[3], bbox[2])
window = cv2.rectangle(img.copy(), start_point, end_point, (0, 255, 0), 10)

plt.imshow(window, cmap='gray')
plt.show()
```

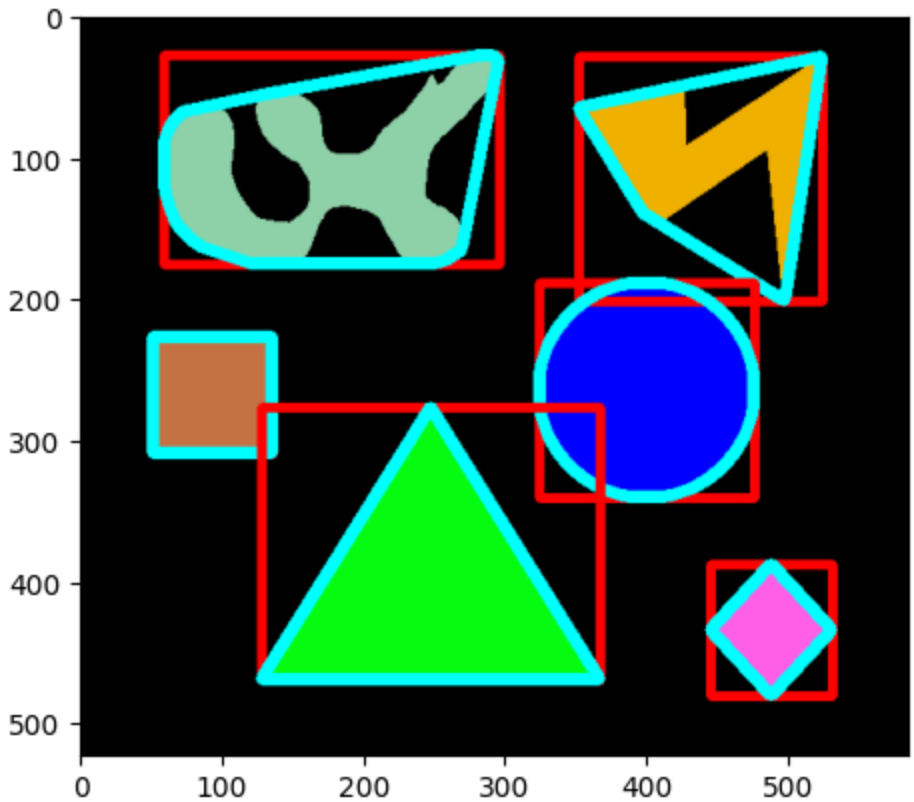
```
(28, 59, 176, 296)
```



In [26]:
```
# Y la convex hull image
from skimage.morphology import convex_hull_image

convex_image = convex_hull_image(objeto)
convex_image = convex_image.astype('uint8')
# plt.imshow(convex_image, cmap='gray')
# plt.show()

conts,_ = cv2.findContours(convex_image, cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE) #
imagen = cv2.drawContours(img.copy(), conts, -1, (200,0,0), 5) # Dibujamos los cont

plt.imshow(imagen, cmap='gray')
plt.show()
```



In [27]:
```
# Repetimos el proceso completo para todos los objetos de la imagen
imagen = img.copy()
for i in range(1, num+1):
    objeto = lab == i
    objeto = objeto.astype('uint8')

    # bbox
    prop = regionprops(objeto)
    bbox = prop[0].bbox
    imagen = cv2.rectangle(imagen, (bbox[1], bbox[0]), (bbox[3], bbox[2]), (255, 0,

    # covex hull
    convex_image = convex_hull_image(objeto)
    convex_image = convex_image.astype('uint8')
    conts,_ = cv2.findContours(convex_image, cv2.RETR_TREE, cv2.CHAIN_APPROX_SIMPLE
    imagen = cv2.drawContours(imagen, conts, -1, (0,255,255), 8)

plt.imshow(imagen, cmap='gray')
plt.show()
```



In [28]:
```
# Vamos a extraer diferentes descriptores geométricos que nos ayudarán a caracteriz
# pip install tabulate
from tabulate import tabulate
import math

new_lab, new_num = label(mask, return_num=True)

# Extraemos las característicias geométricas
headers = ['','Raro', 'Rayo', 'Círculo', 'Cuadrado', 'Triángulo', 'Rombo']
A,BB,CA,E,ED,EX,MA,MiA,OR,P,S,CO,R = ['area'], ['bbox_area'], ['convex_area'], ['ec
['extent'], ['major_axis'], ['minor_axis'], ['orientation'], ['perimeter'], ['solid

for i in range(1,new_num+1):
    objeto = new_lab == i
    prop = regionprops(objeto.astype(np.uint8))

    A.append(np.round(prop[0].area, 4))
    BB.append(np.round(prop[0].bbox_area, 4))
    CA.append(np.round(prop[0].convex_area, 4))
    E.append(np.round(prop[0].eccentricity, 4))
    ED.append(np.round(prop[0].equivalent_diameter, 4))
    EX.append(np.round(prop[0].extent, 4))
    MA.append(np.round(prop[0].major_axis_length, 4))
    MiA.append(np.round(prop[0].minor_axis_length, 4))
    OR.append(np.round(prop[0].orientation, 4))
    P.append(np.round(prop[0].perimeter, 4))
    S.append(np.round(prop[0].solidity, 4))
    CO.append(np.round(4*math.pi*prop[0].area/prop[0].perimeter**2, 4))
    R.append(np.round(prop[0].area/prop[0].bbox_area, 4))
```

```
my_data = [tuple(A), tuple(BB), tuple(CA), tuple(E), tuple(ED), tuple(EX), \
          tuple(MA), tuple(MiA), tuple(OR), tuple(P), tuple(S), tuple(CO), tuple(R)

print(tabulate(my_data, headers=headers))
```

```
                      Raro        Rayo     Círculo    Cuadrado    Triángulo      Rom
bo
--------------  ----------  ----------  ----------  ----------  -----------  -------
--
area            16198       10282       18344        7055        23119       4018
bbox_area       35076       29756       23104        7055        46127       7998
convex_area     27405       16959       18488        7055        23307       4102
eccentricity        0.8556      0.7941      0           0.2157       0.371      0.38
66
equiv_diameter    143.61      114.418     152.828      94.7771     171.569     71.52
54
extent              0.4618      0.3455      0.794       1            0.5012     0.50
24
major_axis        268.45      186.995     152.828      98.1428     195.774     76.19
69
minor_axis        138.975     113.644     152.828      95.8332     181.801     70.27
35
orientation        -1.4236      1.5298     -0.7854      1.5708       1.5708     0
perimeter        1029.77      736.02      503.245     332          717.962    255.58
8
solidity            0.5911      0.6063      0.9922      1            0.9919     0.97
95
compactness         0.1919      0.2385      0.9102      0.8043       0.5636     0.77
29
rectangularity      0.4618      0.3455      0.794       1            0.5012     0.50
24
```

```
C:\Users\erick\AppData\Local\Temp\ipykernel_31976\4149012655.py:18: FutureWarning: `
RegionProperties.bbox_area` is deprecated starting in version 0.26 and will be remov
ed in version 2.0. Use `RegionProperties.area_bbox` instead.
  BB.append(np.round(prop[0].bbox_area, 4))
C:\Users\erick\AppData\Local\Temp\ipykernel_31976\4149012655.py:19: FutureWarning: `
RegionProperties.convex_area` is deprecated starting in version 0.26 and will be rem
oved in version 2.0. Use `RegionProperties.area_convex` instead.
  CA.append(np.round(prop[0].convex_area, 4))
C:\Users\erick\AppData\Local\Temp\ipykernel_31976\4149012655.py:21: FutureWarning: `
RegionProperties.equivalent_diameter` is deprecated starting in version 0.26 and wil
l be removed in version 2.0. Use `RegionProperties.equivalent_diameter_area` instea
d.
  ED.append(np.round(prop[0].equivalent_diameter, 4))
C:\Users\erick\AppData\Local\Temp\ipykernel_31976\4149012655.py:23: FutureWarning: `
RegionProperties.major_axis_length` is deprecated starting in version 0.26 and will
be removed in version 2.0. Use `RegionProperties.axis_major_length` instead.
  MA.append(np.round(prop[0].major_axis_length, 4))
C:\Users\erick\AppData\Local\Temp\ipykernel_31976\4149012655.py:24: FutureWarning: `
RegionProperties.minor_axis_length` is deprecated starting in version 0.26 and will
be removed in version 2.0. Use `RegionProperties.axis_minor_length` instead.
  MiA.append(np.round(prop[0].minor_axis_length, 4))
C:\Users\erick\AppData\Local\Temp\ipykernel_31976\4149012655.py:29: FutureWarning: `
RegionProperties.bbox_area` is deprecated starting in version 0.26 and will be remov
ed in version 2.0. Use `RegionProperties.area_bbox` instead.
  R.append(np.round(prop[0].area/prop[0].bbox_area, 4))
```

5. Github

https://github.com/Erick-305/machine-learning.git